

Ê PHASOR TEMPLATES (Air: ε_r=μ_r=1)

Wavenumber per medium *nonmag.*

$k_i = \frac{\omega\sqrt{\epsilon_{r,i}}}{c}$ (rad/m) $\Rightarrow k_2 = k_1\sqrt{\epsilon_{r,2}/\epsilon_{r,1}}$

Direction table $u \in \{x, y, z\} = \text{axis perp. to boundary}$

Inc dir	Inc	Refl	Trans
+û	e^{-jk_1u}	e^{+jk_1u}	e^{-jk_2u}
-û	e^{+jk_1u}	e^{-jk_1u}	e^{+jk_2u}

+û direction ⇔ Med 2 at u ≥ 0. Reflected sign flips, transmitted matches incident.

General template (incident has pol. ê, amplitude a₀):

$\tilde{\mathbf{E}}^i = a_0\hat{e}e^{\mp jk_1u}$ (V/m) (incident wave, Med 1)

$\tilde{\mathbf{E}}^r = \Gamma a_0\hat{e}e^{\pm jk_1u}$ (V/m) (reflected wave, Med 1)

$\tilde{\mathbf{E}}^t = \tau a_0\hat{e}e^{\mp jk_2u}$ (V/m) (transmitted wave, Med 2)

(lossy med 2: replace $e^{\mp jk_2u}$ with $e^{\mp \alpha_2u}e^{\mp j\beta_2u}$)

Total in medium 1: $\tilde{\mathbf{E}}_1 = \tilde{\mathbf{E}}^i + \tilde{\mathbf{E}}^r$.

ê by polarization *plug into templates above*

Pol.	ê	Note
Lin. x̂	â	E _y = 0
Lin. ŷ	ŷ	E _x = 0
Lin. at 45°	(â + ŷ)/√2	in phase
RHC	â - jŷ	δ = -π/2
LHC	â + jŷ	δ = +π/2
General	a _x â + a _y e ^{jδ} ŷ	elliptical

Γ, τ act on ê unchanged. Only swap the polarization, not the formulas.

TABLE 8-2 — Γ, τ, R, T SUMMARY

Med. 1 (z < 0) incident side; Med. 2 (z ≥ 0) transmitted.

	Normal⊥ pol (θ _i =0)	pol
Γ refl. wave	$\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$	$\frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$
τ trans. wave	$\frac{2\eta_2}{\eta_2 + \eta_1}$	$\frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$
τ rel.	$\tau = 1 + \Gamma$	$\tau_{\perp} = 1 + \Gamma_{\perp}$
R refl. pwr	$ \Gamma ^2$	$ \Gamma_{\perp} ^2$
T trans. pwr	$ \tau ^2 \frac{\eta_1}{\eta_2}$	$ \tau_{\perp} ^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$
R+T total	$T = 1 - R$	$T_{\perp} = 1 - R_{\perp}$

Notes: $\sin \theta_t = \sqrt{\mu_1 \epsilon_1 / \mu_2 \epsilon_2} \sin \theta_i$; *nonmag.* ⇒ $\eta_2 / \eta_1 = n_1 / n_2$.

Intrinsic impedance η_i *plug into table above*

$\eta_i = \sqrt{\frac{\mu_r}{\epsilon_r}} \eta_0 \xrightarrow{\mu_r=1} \frac{\eta_0}{\sqrt{\epsilon_{r,i}}} \text{ (}\Omega\text{)}$ $\eta_0 = 120\pi \approx 377 \Omega$

Default $\mu_r = 1$ (air, dielectric). Use $\mu_r \neq 1$ only if problem says “ferromagnetic”/iron/ferrite/etc.

Low-loss correction $\eta_c \approx \sqrt{\frac{\mu}{\epsilon}} \left(1 + j \frac{\sigma}{2\omega\epsilon'}\right) \xrightarrow{\text{drop } j} \frac{\eta_0}{\sqrt{\epsilon_r}}$

For Γ/τ just use $\eta_0/\sqrt{\epsilon_r}$. The j term is for |η_c|, θ_η. See LOSSY MEDIA — 5 CASES for all regimes.

POWER REFLECTION & TRANSMISSION

% reflected *always*

$\%P_r = 100|\Gamma|^2$

% transmitted *η-ratio matters!*

$\%P_t = 100|\tau|^2 \frac{\eta_1}{\eta_2}$

Check: %P_r + %P_t = 100.

BOUNDARY

η₁, η₂ — imp. (Ω)
Γ, τ, R, T — coeffs (unitless)
τ = 1 + Γ; R = |Γ|²
k_i = ω√ε_{r,i}/c (rad/m)
α₂ (Np/m), β₂ (rad/m): lossy
z = 0 — boundary plane (m)
η₀ = 120π ≈ 377 Ω

SNELL / OBLIQUE

θ_i, θ_t — angles from normal
n_i = √ε_{r,i} — index of refr.
t_i — slab thickness (m)
d — lateral disp. (m)
plane of inc. = plane of k̂+normal
⊥: **E** ⊥ this plane
||: **E** || this plane

WAVEGUIDE

a, b — wide, narrow dim. (m);
a > b
m, n — mode indices (int.)
f_{mn} — cutoff (Hz)
β — prop. const (rad/m)
Z_{TE}/Z_{TM} — wave imp. (Ω)
E_x, H_y — field comps (V/m, A/m)
η = η₀/√ε_r

VELOCITIES

u_{p0} = c/√ε_r — bulk phase vel.
u_p = u_{p0}/√1 - (f_c/f)²
u_g = u_{p0}√1 - (f_c/f)²
u_pu_g = u_{p0}²
t = L/u_g — travel time
c = 3 × 10⁸ m/s

POWER & UNITS

%P_r = 100|Γ|²
%P_t = 100|τ|²η₁/η₂
%P_r + %P_t = 100
σ (S/m), ε (F/m), μ (H/m)
ε₀ = 8.85 × 10⁻¹² F/m
μ₀ = 4π × 10⁻⁷ H/m
1/(36π) × 10⁻⁹ ≈ ε₀

SNELL'S LAW — OBLIQUE INCIDENCE

Single boundary *angles from normal*

$\sin \theta_2 = \sin \theta_1 \sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}} = \sin \theta_1 \frac{n_1}{n_2}$

Multilayer chain *stack of slabs 1 → 2 → 3 → 4*

$\sin \theta_{i+1} = \sin \theta_i \sqrt{\frac{\epsilon_{r,i}}{\epsilon_{r,i+1}}}$

Air-slabs-air shortcut *outer media identical*

θ_{exit} = θ_{incident}

Snell only sets angles. Reflection/transmission magnitudes need Fresnel coefficients (parallel vs perpendicular).

LATERAL DISPLACEMENT

Beam offset through N slabs thickness t_i, refracted angle inside slab i

Slab-indexed θ_i = angle inside slab i; t_i = slab thickness (m)

$$d = \sum_{i=1}^N t_i \tan \theta_i$$

The angle (θ_i) in the formula is always the one the ray makes inside that slab of height (t_i).

LOSSY MEDIA — 5 CASES

Loss tangent $\epsilon' = \epsilon_r \epsilon_0$; $\epsilon_0 = 8.854 \times 10^{-12}$ F/m

$\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega \epsilon_r \epsilon_0}$

Type	σ/ωε	η _c , α, β
Lossless (σ=0)	0	η = η ₀ /√ε _r exact α = 0, β = ω√ε _r /c
Low-loss	< 10 ⁻²	η _c ≈ $\frac{\eta_0}{\sqrt{\epsilon_r}} \left(1 + j \frac{\sigma}{2\omega\epsilon'}\right)$ α ≈ $\frac{\sigma\eta_0}{2\sqrt{\epsilon_r}}$, β ≈ $\frac{\omega\sqrt{\epsilon_r}}{c}$
Quasi-cond.	10 ⁻² –10 ²	exact: (η/√X)e ^{jθ} η see below
Good cond.	> 10 ²	η _c ≈ (1+j)√πfμ/σ α ≈ β ≈ √πfμσ
Magnetic	any	keep full √μ _r /ε _r η ₀

For Γ/τ: drop j correction; use η₀/√ε_r. Magnetic: keep μ_r.

Low-loss quick params σ/(ωε) ≪ 1, μ_r = 1

α ≈ $\frac{\sigma\eta_0}{2\sqrt{\epsilon_r}}$ (Np/m), β ≈ $\frac{\omega\sqrt{\epsilon_r}}{c}$ (rad/m), η_c ≈ $\frac{\eta_0}{\sqrt{\epsilon_r}}$ (Ω)

Exact (quasi-cond) $X = \sqrt{1 + (\epsilon''/\epsilon')^2}$

α = ω√ $\frac{\mu\epsilon'}{2}$ √X-1, β = ω√ $\frac{\mu\epsilon'}{2}$ √X+1

θ_η = $\frac{1}{2} \arctan \frac{\sigma}{\omega\epsilon'}$

RECTANGULAR WAVEGUIDE — MODES

TE mode E_z = 0, H_z ≠ 0

TM mode H_z = 0, E_z ≠ 0

Dimensions a × b with a > b along x̂, ŷ; wave in ẑ.

E_x form (TE) *read m, n from arguments*

$E_x \propto \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) \sin(\omega t - \beta z)$

coef. on x = mπ/a; coef. on y = nπ/b.

Identify TE_{m,n} vs TM_{m,n} *cos/sin pattern of E_x is identical for both!*

1. Read the problem statement (e.g. “a TE wave” → TE).

2. If m = 0 or n = 0 in the mode label ⇒ must be TE (TM forbids 0).

3. Source field H_z(x, y) given ⇒ TE. Source field E_z(x, y) given ⇒ TM.

TE allows at least one of m, n ≥ 1 (other can be 0). TM needs both m, n ≥ 1. That's why TE₁₀ exists but TM₁₀ doesn't.

TABLE 8-3 — FULL FIELD COMPONENTS

All fields carry an e^{-jβz} factor (omitted below). k_c² = (mπ/a)² + (nπ/b)².

TE mode *generator: $\tilde{H}_z$*

$\tilde{H}_z = H_0 \cos \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$

$\tilde{E}_x = \frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b}\right) H_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$

$\tilde{E}_y = -\frac{j\omega\mu}{k_c^2} \left(\frac{m\pi}{a}\right) H_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$

$\tilde{E}_z = 0, \tilde{H}_x = -\tilde{E}_y/Z_{TE}, \tilde{H}_y = \tilde{E}_x/Z_{TE}$

TM mode *generator: $\tilde{E}_z$*

$\tilde{E}_z = E_0 \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$

$\tilde{E}_x = -\frac{j\beta}{k_c^2} \left(\frac{m\pi}{a}\right) E_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$

$\tilde{E}_y = -\frac{j\beta}{k_c^2} \left(\frac{n\pi}{b}\right) E_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$

$\tilde{H}_z = 0, \tilde{H}_x = -\tilde{E}_y/Z_{TM}, \tilde{H}_y = \tilde{E}_x/Z_{TM}$

TEM (plane wave) for reference: $\tilde{E}_x = E_0 e^{-j\beta z}, \tilde{H}_y = \tilde{E}_x/\eta, f_c = N/A, k = \omega\sqrt{\mu\epsilon}$.

CUTOFF FREQUENCY

f_{mn} (a.k.a. f_c for that mode) *m, n integers ≥ 0 (not both zero)*

$f_c \equiv f_{mn} = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$ (Hz)

Common cutoffs (a > b)

Mode	f _{mn}
u _{p0}	c/√ε _r (hollow: u _{p0} = c)
TE ₁₀	f ₁₀ = u _{p0} /(2a) (dominant)
TE ₀₁	f ₀₁ = u _{p0} /(2b)
TE ₂₀	f ₂₀ = u _{p0} /a = 2f ₁₀
TE ₁₁ , TM ₁₁	f ₁₁ = $\frac{u_{p0}}{2} \sqrt{1/a^2 + 1/b^2}$

Propagation requires f > f_{mn}. In Z_{TE}/Z_{TM} and u_g, f_c = f_{mn} of the excited mode.

ε_r, FROM DISPERSION

Given ω, β, mode (m, n) *result is unitless*

$\epsilon_r = \frac{c^2}{\omega^2} \left[\beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \right]$ (unitless)

Inputs: ω (rad/s), β (rad/m), a, b (m), c = 3 × 10⁸ (m/s), m, n (unitless).

WAVE IMPEDANCE & H_y

$Z_{TE} = \frac{\eta}{\sqrt{1 - (f_c/f)^2}}$ (Ω) $Z_{TM} = \eta \sqrt{1 - (f_c/f)^2}$

$\eta = \frac{\eta_0}{\sqrt{\epsilon_r}}$ $H_y = \frac{E_x}{Z_{TE}}$ (TE mode)

Z_{TE} > η always (denominator < 1); Z_{TM} < η.

GROUP VELOCITY & TRAVEL TIME

$u_g = u_{p0} \sqrt{1 - (f_{mn}/f)^2}$ (m/s)

$u_p = \frac{u_{p0}}{\sqrt{1 - (f_{mn}/f)^2}}$ (m/s) ⇒ $u_p u_g = u_{p0}^2$ (m²/s²)

t = L/u_g (s) travel time through guide of length L

Higher modes have lower u_g ⇒ they arrive later. Group delay spread = sum over excited modes.